

Utilization of a robotic arm in wind tunnel for the study of gust energy harvesting

1. Context

It is today well known that the UAV -Unmanned Aerial Vehicles- development is mainly restrained because of the battery weaknesses. The weight and space available reduce substantially the possibilities so that the autonomy of the device is quite low, which is why the research has been proactive in investigating the ways to retrieve the energy available in the atmosphere. This idea comes from the nature and more especially birds, that have demonstrate the ability to make use of this energy in order to fly "at low cost" in some configurations. For instance, albatross have develop the capacity to use the oceanic boundary layer to fly with neutral energy cycles. To sum up, the atmosphere provides three sources of energy that can be used by something flying : vertical air motion (thermal instabilities,...), non-stochastic spatial wind gradients (shear layers) and stochastic spatial or temporal gradients (gusts) [1]. In our context, it is this last one we are interested in. The basics of the gust energy harvesting is well explained in [2] and is easy to understand for a case of longitudinal flight with a gust coming from upward/downward. As it can be seen in Figure 1, whether the gust comes from upward or downward, lift gains a component of thrust given the fact that it is always perpendicular to the local wind. It is this component we are looking to add to the thrust auto-generated by the UAV. The principle seems quite simple but it needs to be reminded that UAVs are meant to be used in city environment, which implicate much more complicated gust field than a simple longitudinal sinusoidal gust. Moreover, it is obvious that in order to benefit at maximum from the energy harvesting, it will require an additional use of the UAV's actuators, which means an additional energy cost. In consequen-

ce, we have to be aware that harvesting energy from gusts is good but gain more than we expend is better.

2. State of the art

The first studies have been targeted the easiest/closest to nature aspects of the problem. Indeed, the neutral energy flight cycle of the albatross have been deeply investigated [3] [4] as well as techniques for static soaring (i.e. finding out which speed is the better for a given wind field with no gradient). These studies were necessary to lay a groundwork for later dynamic soaring studies by establishing for example the maximum speed with no net altitude loss [5] . For the major part, the dynamic soaring studies are simulation studies with a common structure and distinguish themselves by the model built for the simulation. It starts with the choice of the wind model, i.e. deciding how to recreate the outside atmospheric environment. Then it is necessary to do the same for the UAV and the system of flight dynamic. Some studies even anticipate the issue of the sensors [2]. Once the system and the environment are well defined comes the law control design based on maximizing energy and finally, simulations are run.

The wind model relies mainly on providing a Power Spectral Density (PSD) able to reproduce real gusts. The gust field is generally built by superposing several sinusoids whose amplitude and frequency matches with their relative contribution to gust intensity and PSD. The phase of each sinusoid is determined randomly. Today two PSD spectra are considered quite accurate for this type of study. The one of Von Karman or Dryden has been judged representative enough of natural turbulence and is used in the works of Patel and Kroo [2] or Langelaan [1] for example. Others

use the Kailman and Finnigan spectrum [6], developed on measurements over flat homogeneous terrain, more representative of the UAVs environment.

The two different paths for the UAV model are either choose a mass point model [1] [2] or a multi-points mass model. Indeed, in [6], the UAV model is made more complex by adding two more points on the wings. In consequence, it transforms the original 2D model in a 3D model which requires adding roll and yaw moments in order to take into account the difference of local wind speed on the different parts of the wing.

Designing the law control requires first to exploit dynamic flight equations in the objective of determine which parameters influence the energy fluctuation of the UAV. However, optimizing naively mass energy or energy per distance will obviously lead to order a maximum speed, i.e. lose altitude, etc... It is then required to determine an optimal airspeed, lift coefficient and flight path angle for each time step with some constraints ensuring that the UAV is able to fly as long as possible. Distinctions are made on the output of the law control, which can be the lift coefficient [2], the elevator deflection [1], the elevator and flaps deflection [6]. Once it is done, simulations are run with a set of generated wind model to find out which gains of the controller ensure the tougher law control.

Respecting experimentation, and contrary to simulation, the literature is quite poor. In fact, one of the only study dealing with real tests of the law control is [2]. For this purpose, a prototype of UAV has been built and instrumented in order to fly outdoor.

3. Prospects

The very interest of the proposed subject is the possibility to test and recreate a turbulent environment thanks to the SABRE wind tunnel and the robotic arm. In fact, in [2], Patel & Kroo did obtain concluding results (a mean of 46.22 % energy improvement compared to a standard UAV) but they were constrained in their conclusions by the non-repeatability of a flight due to outdoor conditions so that they couldn't test the influence of geometrical parameters for example. Indeed, it is suggested in [6] that some models of UAV might be more adapted to dynamic soaring than others. To sum up, what can bring a wing tunnel is the ability to verify more easily the efficiency or not of some models (multipoints mass UAV, etc) but mostly lay a background for experimentation in this field as it is easy control the phenomena, to instrument UAV and thus to obtain data.

4. References

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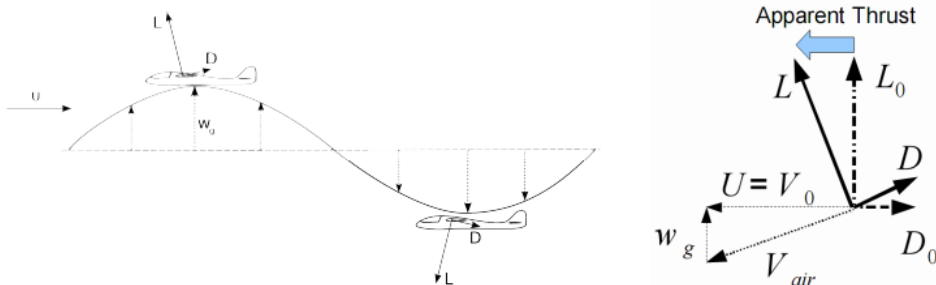


Figure 1: Fundamental concept of gust energy harvesting [2]